Junsang Park

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Research Interests

Protein Design and Engineering: Modeling for protein design and predicting interactions
AI-driven drug discovery: Machine Learning for chemical representation, protein–ligand recognition

Education

Seoul National University (SNU), Seoul, Korea — Mar. 2023 – Present
B.S. in Chemical and Biological Engineering · Minor in Computer Science and Engineering

- GPA: **4.26**/4.30 (overall) · **4.25**/4.30 (major) · **4.25**/4.30 (minor)
- Relevant Courses: Deep Learning(Graduate)· Computer Vision · Molecular Biochemical Engineering

Daejeon Science High School for the Gifted (DSHS), Daejeon, Korea — Mar. 2020 – Feb. 2023
Specialized high school for gifted students in science and mathematics

- Graduated with Academic Excellence Award

Publications

† Equal contribution; * Corresponding author

[J1] S. Lee†, **J. Park**†, and Y. Lee*. A web-based interactive toolkit for ternary phase equilibrium and liquid-liquid extraction. To be submitted to *Education for Chemical Engineers* (2026).

Research Experience

Computational Biology Laboratory, SNU, Seoul, Korea
Undergraduate Researcher, *Advisor: Prof. Chaok Seok* — Feb. 2026 – Present

- Topic: **Structure-based machine learning** for understanding **dynamic biomolecular behavior**
  - Investigating structural determinants of GPCR agonism and antagonism and their coupling to receptor conformational dynamics
  - Initiating a study of the structural and dynamic basis of CBAD–mediated regulation in IL–17RA signaling

Biomedical & Health Informatics Laboratory, SNU, Seoul, Korea
Undergraduate Researcher (UROP), *Advisor: Prof. Sun Kim* — Sep. 2025 – Jan. 2026

- Topic: Protein language model-guided **antibody engineering** and **mutation effect prediction**
  - Contributed to the development of an ESM-based protein language model pipeline to propose sequence mutations via ensemble voting

Selected Projects & Competitions

CBPL–kit: Interactive Hunter–Nash LLE simulator Project Website
Lead Developer — Mar. 2026 – Present

- Led design and development of an interactive platform for visualizing multistage liquid–liquid extraction using the Hunter–Nash method
- Designed interactive ternary-diagram visualizations and stage-by-stage calculations to help students connect phase-equilibrium theory with extraction process design
- Scheduled for integration into an undergraduate chemical engineering course at Seoul National University in Fall 2026

SNU FastMRI challenge, SNU Electrical and Computer Engineering
Participant — Project in Progress

- Developing neural networks for accelerated MRI reconstruction from undersampled k-space data

Chemical Engineering Process Design Competition,The Korean Institute of Chemical Engineers
Participant — Project in Progress

- Developing ML models to predict the kinetics of cumene production reaction

Selected Awards & Honors

Kwanjeong Scholarship, Kwanjeong Foundation — Mar. 2025 – Present
Selected as one of approximately 100 domestic scholars annually from university-nominated candidates
Awarded $2700 per semester (total 4 semesters)

Presidential Science Scholarship, Korea Student Aid Foundation — Mar. 2023 – Present
National scholarship awarded to outstanding students in STEM field
Awarded $3700 per semester (total 8 semesters); Covering full tuition and an academic stipend

Special Award, 67th National Science Fair, National Science Museum — 2021
Topic: *Sustained Use of Self-Healing Road-Marking Beads Containing Microorganisms*
Selected from 2,300 submissions nationwide; One of Korea's longest-running national science competitions

Academic Service & Activities

Conference Volunteer, 43rd International Conference on Machine Learning (ICML 2026) — Jul. 2026
Assisted attendee registration and on-site conference operations at an international machine learning conference

STEM: SNU Tomorrow's Engineers Membership, College of Engineering, SNU — 2025 – Present

- **Director of Recruitment:** Led member recruitment, selection logistics for new STEM members
- **Director of Volunteer Service:** Coordinated student volunteering programs
- **STEM Vision Scholarship:** Supported the selection process of five scholarship recipients; Each awarded $1000
- **Mentor, STEM Vision Fair:** Mentored freshman and junior engineering students on scholarship, academic pathways, and AI–driven drug discovery research

CES 2026 & Silicon Valley Global Exploration Program, SNU — Sep. 2025 – Feb. 2026
Selected as a participant to explore emerging technologies and the Silicon Valley innovation ecosystem

Peer Tutoring Program, SNU — 2024,2025

- **Major course Peer Tutoring Program**
  Tutor, *Organic Chemistry II*(2025)
- **Basic Course Peer Tutoring Program**
  Tutor, *Introductory Physics I*(2024), *Fundamentals of Physics II*(2025)

Skills

Programming: Python, C++, HTML/CSS, LaTeX
Tools: Git, WSL/Linux, Docker
Computational Biology: PyMOL, UCSF ChimeraX, AlphaFold/ColabFold
Process Simulation: Aveva Process Simulator, Pro/II
Languages: English (Fluent), Korean (Native)